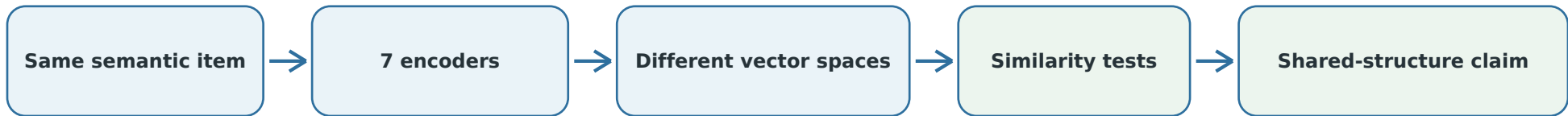


Embedding Spaces, Vectors, Local & Global Similarity

Corrected for the full 7-encoder study: 3 DINO vision spaces + BGE, GPT-2, BERT, and SBERT text spaces



Running example

Image: a cat sitting on a chair. Text: "A cat is sitting on a chair." The raw inputs differ, but they refer to the same underlying item.

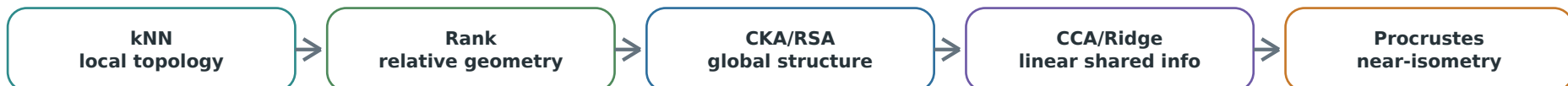
Central question

When two models encode the same item, what survives across their representation spaces: coordinates, direction, local neighbors, distance rankings, global geometry, or only a linearly recoverable subspace?

Key distinction

Same vectors is the strongest and usually wrong requirement. A more realistic hierarchy is: shared local topology -> shared global relations -> shared linear structure -> near-isometry.

Evidence ladder: weaker assumptions on the left, stronger equivalence on the right



1. Space, Vector, Representation Matrix, Ambient Dimension

The basic objects: where vectors live, how many coordinates they have, and what the data actually occupies

Vector

A single embedding is one point with d coordinates. Example: x_{cat} in $\mathbb{R}^{1536}$.

$$x = [x_1, x_2, \dots, x_d] \text{ in } \mathbb{R}^d$$

Representation matrix

For N aligned samples, stack one vector per row. Rows are items; columns are embedding dimensions.

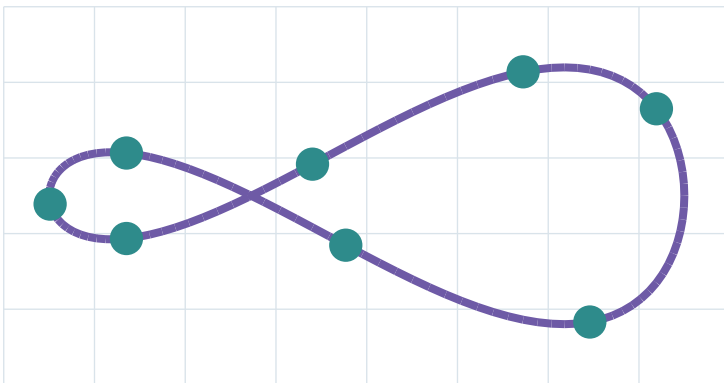
$$X \text{ in } \mathbb{R}^{(N \times d)}$$

Ambient vs intrinsic dimension

A model may output 1536 coordinates, while the meaningful data occupy a much smaller curved manifold inside that ambient space.

$$\text{intrinsic_dim}(M) \ll \text{ambient_dim}(\mathbb{R}^d)$$

Visual: a low-dimensional manifold embedded in a larger space



Ambient coordinates can be high-dimensional; the data may follow a much lower-dimension

Cat-chair example

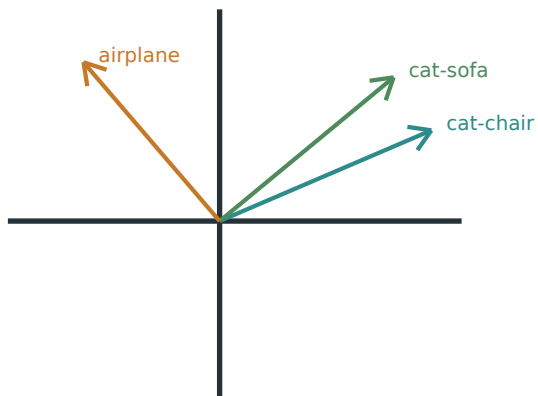
DINOv2-base may encode the image as x_{img} in $\mathbb{R}^{1536}$. BGE may encode its caption as x_{txt} in $\mathbb{R}^{1024}$. These points cannot be compared coordinate-by-coordinate, but both can still encode the same relational facts: cat-like items are near, furniture-related items are near, airplanes are far.

$$x_{\text{img}} \text{ in } \mathbb{R}^{1536} \quad \neq \quad x_{\text{txt}} \text{ in } \mathbb{R}^{1024}$$

2. Direction, Distance, Geometry, and Topology

Four related ideas that are often mixed together

Direction and cosine



$$\cos(x,y) = (x \cdot y) / (||x|| \ ||y||)$$

Cosine compares direction, not absolute position or scale. It only makes direct sense when both vectors are expressed in the same coordinate system.

Geometry

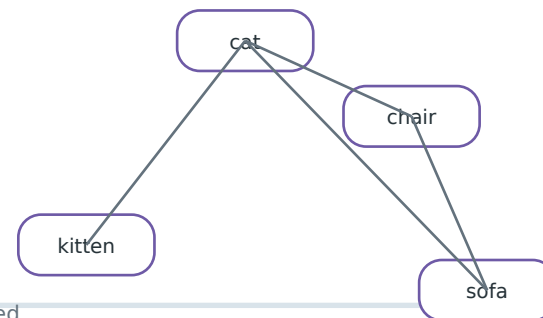
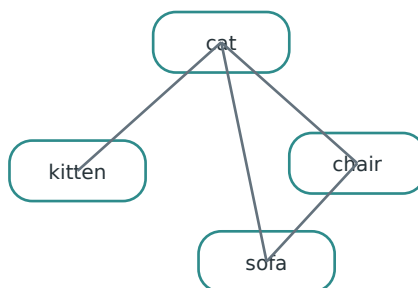
Geometry asks about metric quantities: distances, angles, directions, curvature, and scale. Two spaces can have different coordinates yet preserve geometry under a rotation.

$$d(x,y) = ||x - y||_2$$

Topology

Topology is weaker: it asks which points are connected or neighboring, not their exact metric distances. A nonlinear warp can preserve neighborhoods while changing angles and distances.

Same topology, different geometry



3. Local Similarity: k-Nearest Neighbors (kNN)

A coordinate-free test of whether two models agree on who belongs near whom

What kNN measures

For each item i , find its k nearest neighbors inside each model space. Then compare the two neighbor sets. This measures local relational agreement without requiring equal dimensions or aligned coordinates.

Space A - text/BGE neighborhood

cat on sofa; kitten on chair; pet on furniture; cat indoors; dog on chair

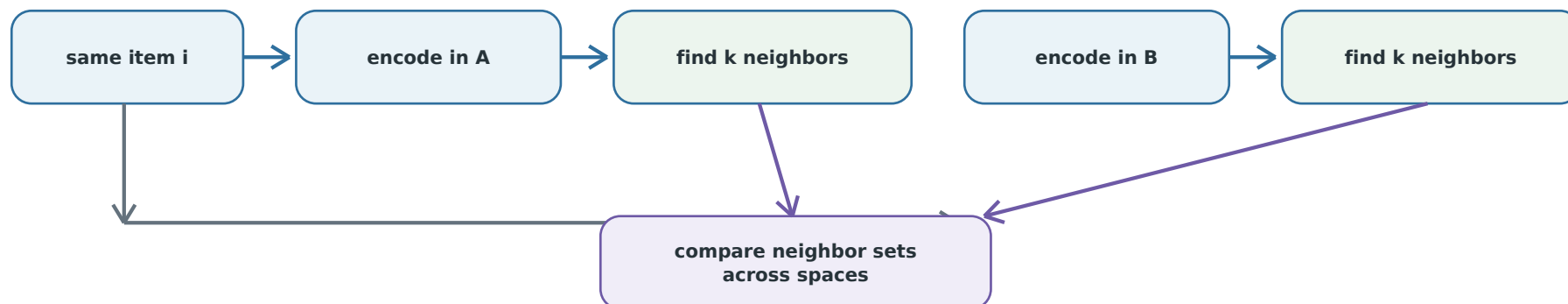
Space B - image/DINO neighborhood

kitten on chair; cat on sofa; cat indoors; pet on furniture; bird on chair

$$\text{Overlap}_k(i) = |N_A^k(i) \cap N_B^k(i)| / k$$

$$\text{Overlap}_{@5} = 4/5 = 0.80$$

Local-topology flow



Why local is scientifically useful

Local neighborhoods often survive nonlinear coordinate changes that destroy exact global metric agreement. This makes kNN a sensitive test for a shared manifold under different charts.

Why test several k values

$k=5$ probes very local structure; $k=500$ probes broader neighborhoods. If agreement is high for small k and decays with k , the spaces share local topology more strongly than global geometry.

Project status

Multi- k neighborhood overlap should be computed across all 21 pairs of the 7-space experiment: DINO-small/base/large, BGE, GPT-2, BERT, and SBERT. It is distinct from retrieval $R@K$, which asks whether the exact paired target is retrieved.

4. Global Similarity: Distances, Rankings, Gram Structure, CKA

Global tests ask whether relationships across many or all samples are organized similarly

Pairwise distance matrix

Build D_A and D_B , where entry (i,j) is the within-space distance between samples i and j . Compare the overall pattern rather than raw coordinates.

$$D_A[i,j] = d(x_i^A, x_j^A)$$

Distance-rank agreement

If one model says cat-chair is closer to cat-sofa than to airplane, and the other preserves the same ordering, rank correlation can be high even when absolute distances differ.

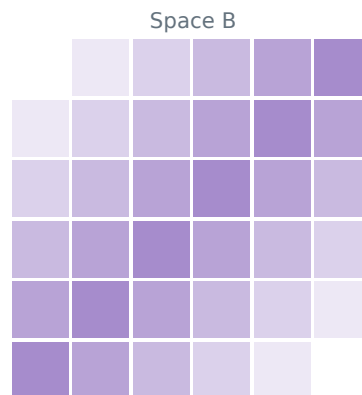
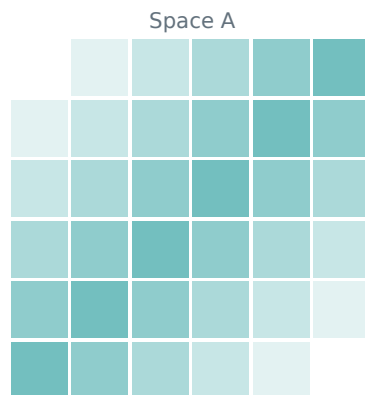
$$\rho = \text{Spearman}(\text{vec}(D_A), \text{vec}(D_B))$$

Gram structure / CKA

CKA compares global representational structure through centered similarity/Gram matrices and is invariant to several coordinate symmetries. It is symmetric, so A vs B equals B vs A .

$$\text{CKA}(X,Y) = ||X^T Y||_F^2 / (||X^T X||_F ||Y^T Y||_F)$$

Visual intuition: local vs global



High local, weak global

Small neighborhoods agree, but long-range distances and angles are warped.

High global agreement

The large-scale similarity pattern is preserved across most sample pairs.

Interpretation ladder

kNN high only -> shared local topology. Rank high -> relative distances are ordered similarly. CKA high -> broad global structure is aligned. None of these alone proves that an embedding can be translated into the other space.

5. Can One Space Be Translated Into Another?

Ridge, CCA, and Orthogonal Procrustes test progressively stronger forms of shared structure

Ridge / linear mapping

Learn W from aligned training rows so that source vectors predict target vectors. If held-out retrieval or R^2 is high, shared information is globally linearly accessible.

$$\hat{Y} = X W$$

$$W = (X^T X + \lambda I)^{-1} X^T Y$$

CCA / shared subspace

CCA does not require one space to predict the full other space. It searches for pairs of directions that are maximally correlated, revealing common latent factors plus private information.

$$u = X a, \quad v = Y b$$

maximize $\text{corr}(u, v)$

Orthogonal Procrustes

Restrict W to a rotation/reflection R . Strong performance means the spaces are nearly the same geometry in different coordinates, rather than merely linearly related.

$$Y \approx X R, \quad R^T R = I$$

$$X^T Y = U \Sigma V^T \rightarrow R = U V^T$$

Translation flow for the cat-chair item



What a miss means

A linear map below the ceiling does not automatically mean the missing structure is local. The gap can also come from nonlinear warping, dimensional bottlenecks, anisotropy, noise, or information absent from the source encoder.

Directional asymmetry

$A \rightarrow B$ and $B \rightarrow A$ are different tasks. With 7 spaces there are 21 symmetric pairs but 42 directed mappings. Asymmetry can reveal different information content or mapping difficulty.

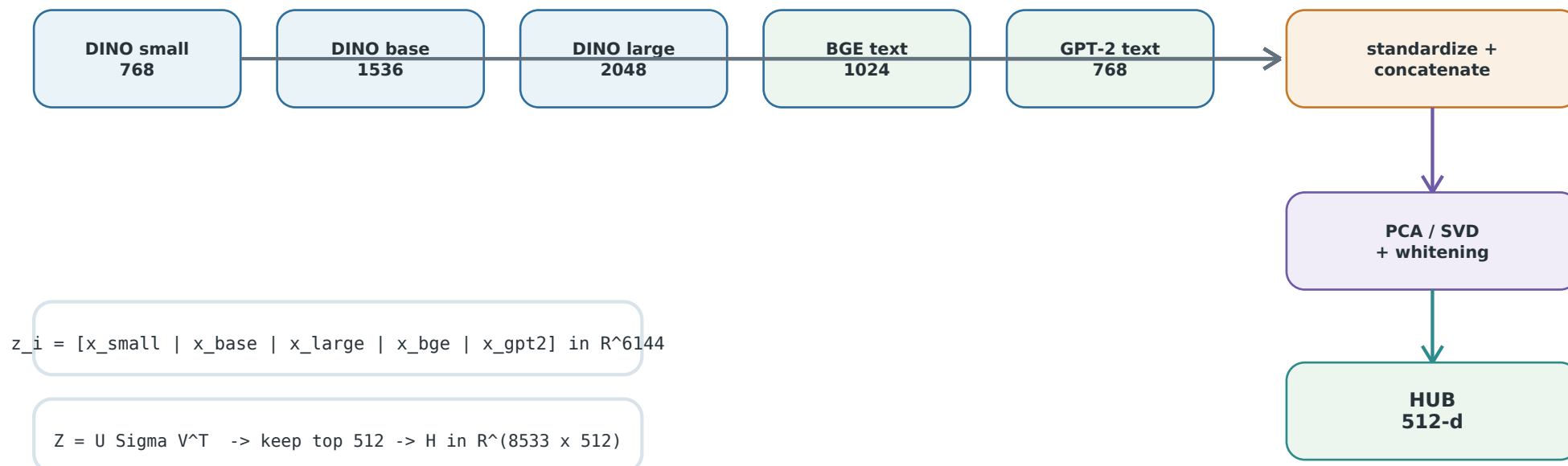
Project-linked tests

Ridge and Procrustes were used in the project. CCA is a complementary diagnostic for shared linear subspaces and is useful when full prediction is too strong a requirement.

6. Shared Hub vs the Full Seven-Encoder Study

Full study = 7 spaces. The documented G1 Hub currently uses 5 arms; BERT and SBERT belong to the full structural study and can be added to the Hub as an extension.

Hub construction from aligned items



Runtime: each encoder learns one map into the Hub



Current G1 meaning: five documented Hub arms map into the shared 512-d language. BERT and SBERT are additional spaces in the full seven-encoder study; they are not silently treated as inputs to the already-built five-arm G1 Hub.

7. Retrieval, Verification, Anisotropy, Hubness, and Collapse

A space can preserve coarse correspondence while losing the fine discrimination needed for exact matching

Retrieval R@K

For a query, rank all target candidates. R@1 asks whether the true paired target is first; R@10 asks whether it appears anywhere in the top 10. This is a hard instance-identification test.

$R@1: \operatorname{argmax}_j \operatorname{sim}(q_i, t_j) = i ?$

Verification AUC

AUC asks a weaker pairwise question: do true pairs tend to score above random/nonmatching pairs? AUC can remain good even when a few hard negatives beat the true target in retrieval.

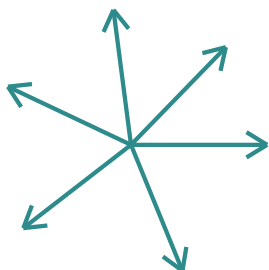
$AUC \sim P(\operatorname{score}_{\text{true}} > \operatorname{score}_{\text{random}})$

Why they can disagree

True pair = 0.80; most negatives = 0.30-0.60; two negatives = 0.83, 0.82. Verification is strong, but R@1 fails.

Geometry failure modes

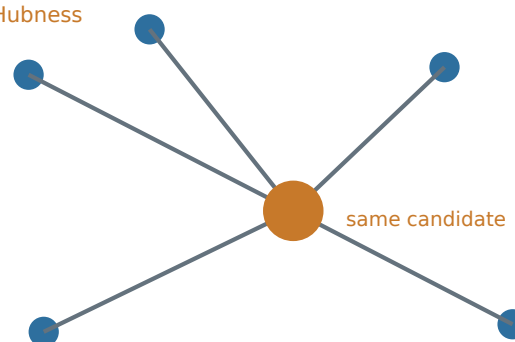
Isotropic



Anisotropic cone



Hubness



strong anisotropy

low effective rank

projection / bottleneck

collapse

R@1 down

8. Why Local Similarity Is Still Platonic

The precise version: different charts can preserve relational invariants without sharing coordinates or a single global metric

Naive claim - too strong

"All capable models converge to the same embedding space." This would suggest directly comparable coordinates or a nearly identical global metric. That is not required by the evidence.

$$Z_A \approx Z_B \quad (\text{too strong})$$

Precise claim

Different models may be different coordinate charts of a partially shared relational manifold. What should be tested is which invariants survive the change of chart.

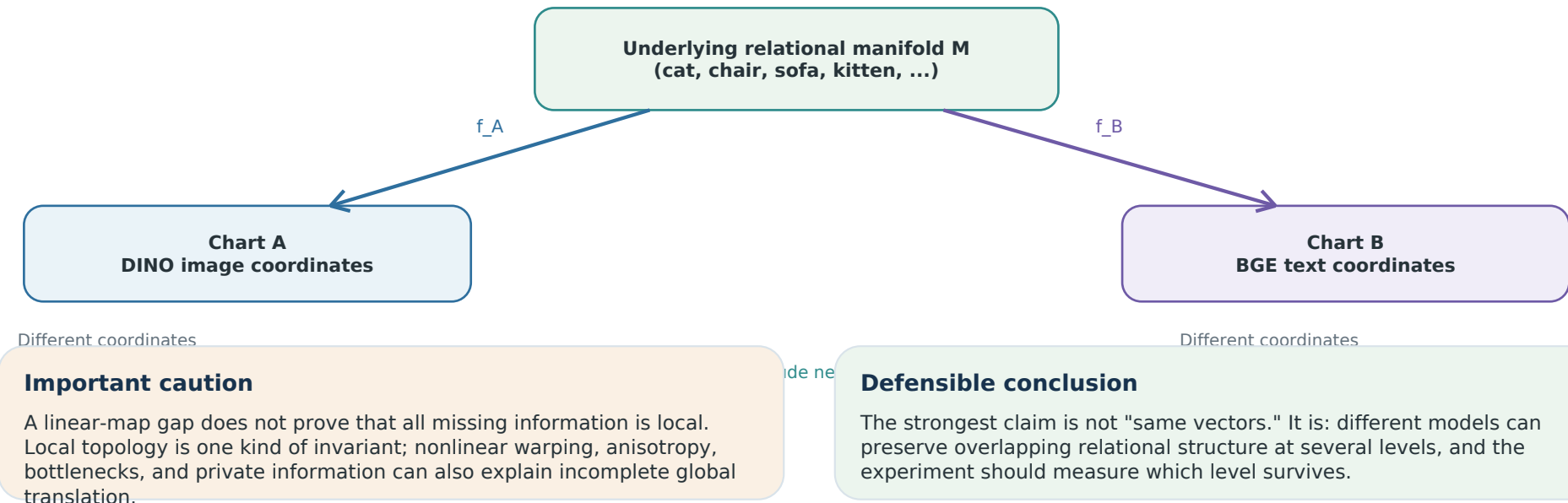
$$\text{exists } M: Z_A = f_A(M), \quad Z_B = f_B(M)$$

Why local counts

If cat-chair has similar nearest neighbors across text and image encoders, both models recovered the same local relational facts even if the coordinates, dimensions, distances, and angles are different.

$$N_A^k(i) \approx N_B^k(i)$$

Chart interpretation



9. Complete Methodology: From Item to Shared-Space Claim

A practical test sequence and how to interpret each result

1. Encode aligned items	Same cat-chair item -> DINO image / BGE text	$X_A \in \mathbb{R}^{(N \times d_A)}; X_B \in \mathbb{R}^{(N \times d_B)}$	aligned rows
2. Diagnose native geometry	Norms, spectrum, effective rank, anisotropy, hubness	SVD + cosine distributions	healthy discrimination?
3. Test local topology	Compare nearest-neighbor sets for the same item rows	kNN overlap@k	local relations survive?
4. Test global structure	Compare distance rankings or Gram/similarity structure	Spearman / RSA / CKA	global relations agree?
5. Test linear accessibility	Fit source -> target on train only; evaluate held-out	Ridge / CCA	linearly recoverable?
6. Test near-isometry	Allow only rotation/reflection after dimension matching	Orthogonal Procrustes	same geometry?
7. Test operational usefulness	Retrieve the exact paired target among all candidates	R@1 / R@5 / R@10	works for search?
8. Build / extend a Hub	Current G1: 5-arm Hub; full study also contains BERT + SBERT	PCA/SVD Hub + model maps	multi-model interoperability
Project-linked tests already used CKA (language layers), Ridge mapping, Orthogonal Procrustes, retrieval R@K, Verification AUC, PCA/SVD Hub construction, Hub-dimension and anisotropy/hubness diagnostics. Complementary / proposed diagnostics Multi-scale kNN overlap across all representation pairs, distance-rank agreement, and CCA provide a cleaner decomposition of local topology, global geometry, and shared linear subspace. One-sentence synthesis Models need not learn the same vectors. The measurable question is which relational invariants survive across their coordinate systems, and whether those invariants are locally, globally, or linearly shared.			

10. The Full Seven-Encoder Experiment - Correct Scope

All seven spaces participate in the representation-comparison study: DINO-small/base/large, BGE, GPT-2, BERT, and SBERT

A. Seven aligned representation spaces

Space	Encoder / modality	Width	Scope
img_small	DINOv2-small / image	768	7-space tests + G1 Hub
img_base	DINOv2-base / image	1536	7-space tests + G1 Hub
img_large	DINOv2-large / image	2048	7-space tests + G1 Hub
txt_bge	bge-m3 / text	1024	7-space tests + G1 Hub
txt_gpt2	GPT-2 / text	768	7-space tests + G1 Hub
txt_bert	BERT / text	768	7-space tests
txt_sberr	SBERT / text	768	7-space tests

7 spaces -> $C(7,2)=21$ symmetric pairs; $7*6=42$ directed mappings

$N=9533$ aligned items = 8533 fit + 1000 held-out eval

Current documented G1 Hub - five arms

DINO-small + DINO-base + DINO-large + BGE + GPT-2 define the existing HUB_512. The supplied G1 construction concatenates 6144 dimensions before PCA/SVD. It should not be described as a seven-arm Hub.

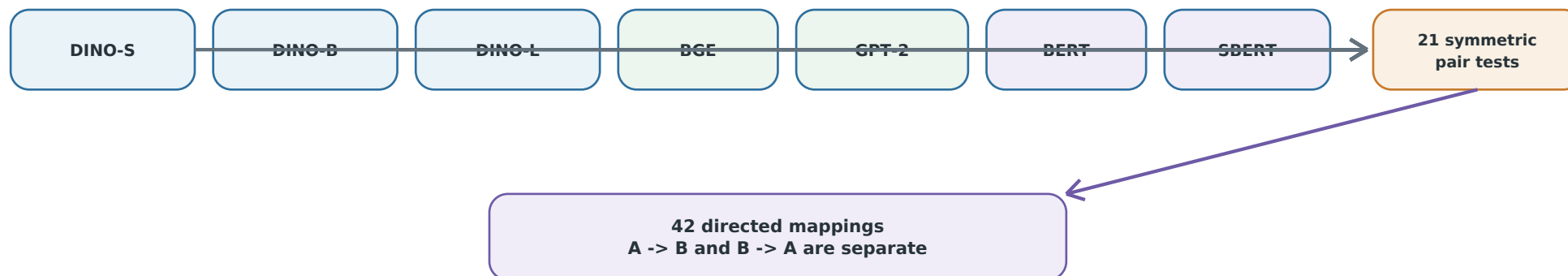
BERT and SBERT - full experiment scope

BERT and SBERT are part of the seven-space comparison and therefore belong in kNN, CKA/RSA, ranking, CCA/Ridge, Procrustes, and 21-pair / 42-direction summaries even though they are not current G1 Hub inputs.

11. Seven-Encoder Structural Flow + Frozen-Hub Extension

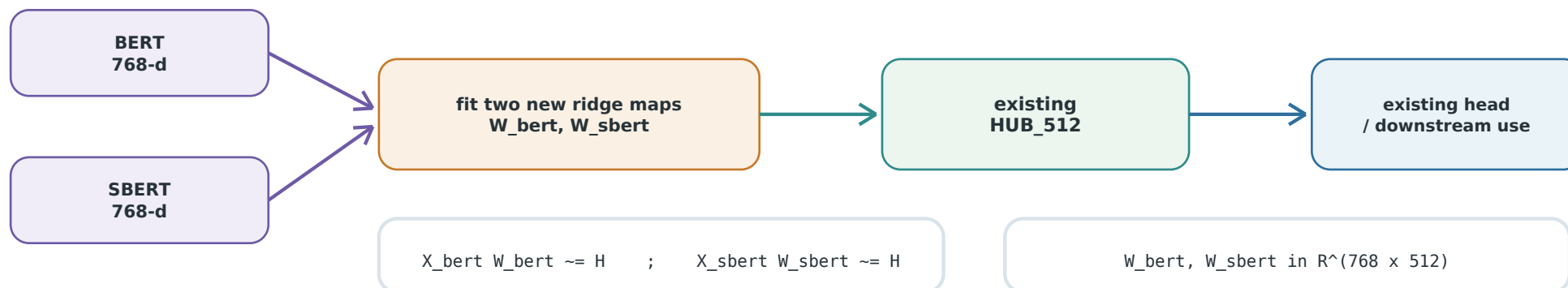
All seven encoders enter the structural tests; BERT and SBERT can be mapped into the already-built five-arm Hub without redefining it

A. Full seven-encoder structural flow



Per pair: local kNN -> rank geometry -> CKA/RSA -> CCA/Ridge -> Procrustes -> held-out directional retrieval.

B. Option 1 - add BERT and SBERT to the frozen existing Hub



This extension teaches BERT/SBERT the old Hub coordinates. It does not change the PCA basis that originally defined HUB_512.

12. Rebuilding the Hub with All Seven Encoders

If all seven encoders should define the shared coordinates themselves, rebuild the concatenated basis and refit dependent maps

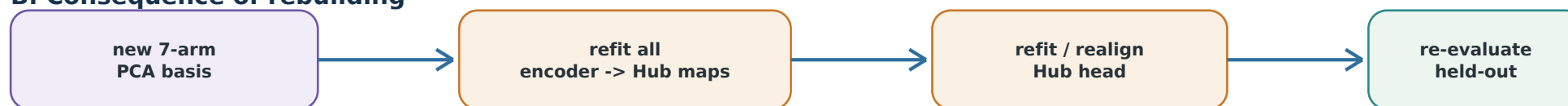
A. Seven-arm Hub construction



$Z_7 = [x_{\text{small}} \mid x_{\text{base}} \mid x_{\text{large}} \mid x_{\text{bge}} \mid x_{\text{gpt2}} \mid x_{\text{bert}} \mid x_{\text{sbert}}]$ in $\mathbb{R}^{(8533 \times 7680)}$

$768 + 1536 + 2048 + 1024 + 768 + 768 + 768 = 7680$

B. Consequence of rebuilding



When frozen-Hub integration is preferable

Use it when the current HUB_512 and head are already validated and the goal is modular compatibility: fit only W_{bert} and W_{sbert} , then test whether their Hub vectors transfer through the existing downstream path.

When seven-arm rebuilding is scientifically interesting

Use it when the research question is whether BERT and SBERT contribute new shared variance and should influence the shared coordinate system itself. Because the Hub axes change, all dependent maps must be revalidated.